

Router IOS Backup and Restore:

Flash memory holds IOS of router/switches

#show flash

copy IOS filename

#copy flash tftp

10.0.0.2

paste filename

enter to go with default filename

Backup done

To Restore

#del flash

paste filename

confirm---enter

#reload

rommon1>?

2>tftpdnld

IP_ADDRESS=10.0.0.1

IP_SUBNET_MASK=255.0.0.0

DEFAULT=GATEWAY=10.0.0.1

TFTP_SERVER=10.0.0.2

TFTP_FILE=paste filename

>tftpdnld

y

>boot

Restore done

Secured Shell

22/tcp Application

is used to access remote device securely.

Steps to configure ssh:

config# hostname Mum

config# ip domain-name abc.com

//setting the domain name

config# Crypto key generate rsa

[512]:enter

config# username john password ssh@123

config# line vty 0 4

config-line)#transport input ssh

config-line)# login local

#end

#write

To verify:

Goto pc

cmd

ssh -l john 10.0.0.1

//ip of router

password:123

DHCP: Dynamic Host Configuration Protocol:

67,68/UDP

assigns IP to clients dynamically

subnet mask, dns address, router address

DHCP works on the process of DORA

ip given by dhcp is a temporary ip

Router:

```
conf t
config#hostname DHCP-Router
config#ip dhcp pool corp
config-dhcp)#domain-name corp.com
config-dhcp)#network 192.168.100.0 255.255.255.0
config-dhcp)#default-router 192.168.100.100
config-dhcp)#dns-server 192.168.100.50
config-dhcp)#end
#sh ip dhcp binding //shows the mapping of ip
given to the dhcp clients
```

```
=====
FTP                File Transfer Protocol                20,21/tcp                Application Layer
                  helps to upload & download files
```

```
=====
DNS:      Domain Name System                53/tcp & UDP
                  resolves domain name to ip address & vice-versa
```

```
=====
SMTP & POP3      SMTP---25/tcp                Outgoing Mails
                  POP3---110/tcp                Incoming mails
                  IMAPv4---143/tcp                Incoming mails
```

Dynamic Routing:

Routing Protocols are responsible for searching available nws, defining best path.

1. less burden on Administrator
2. More overhead on Router's CPU & Memory.
3. Less secure.
4. Suitable for large internetworks.

IRP	ERP
Interior Routing Protocol	Exterior RP
IGP	EGP
Interior Gateway Protocol	Exterior GP
ROUTING is DONE WITHIN A DOMAIN	ROUTING IS DONE BETWEEN A DOMAINS
eg: RIP IGRP EIGRP OSPF IS-IS	BGP (Border Gateway Protocol)

IGP:	AD Value
RIP Routing Information Protocol	120
OSPF: Open Shortest Path First	110
IGRP: Interior Gateway Routing Protocol	100
EIGRP: Enhanced IGRP	Internal-90, External-170
IS-IS: Intermediate System-Intermediate System	115

AD values: V IMP

Administrative Distance Value Its a ranking given by Cisco to each routing protocol that defines the trustworthiness of each routing protocol

Rank can be 0-255 lower the value, highest the trust worthiness

When multiple routing protocols are configured on router, then router will consider routing protocol on the basis of their AD value, and will keep the routing updates of the best Routing protocol in table.

RA	----	RB	-----	RC
Names				AD Values
Static (directly connected)				0
Static (Next Hop)				1
RIP				120
IGRP				100
EIGRP		internal		90
		external		170
OSPF				110
IS-IS				115

Types of Dynamic Routing:

- 1.DVRP: Distance Vector Routing Protocol
- 2.LSRP:Link State RP
- 3.Hybrid (Advanced DVRP)--> (DVRP+LSRP)

RIP, IGRP
OSPF, IS-IS
EIGRP

Working of DVRP: each router exchanges routing table infn with each other periodically.

```

R1 RIP      -----
R3 RIP
  c 10.0.0.0 fa0/0
c 50.0.0.0 fa0/0
  c 20.0.0.0 s0/0
c 40.0.0.0 s0/1
  R (R2) 30.0.0.0
R (R2) 10.0.0.0
  R (R2) 40.0.0.0
R (R2) 20.0.0.0
  R (R2) 50.0.0.0
R (R2) 30.0.0.0

R2 30.0.0.0
40.0.0.0
10.0.0.0/8
20.0.0.0/8
10.0.0.0/30
20.0.0.0/30

R2 RIP      -----
R2 RIP
  c 30.0.0.0 fa0/0
  c 20.0.0.0 s0/0
c 40.0.0.0 s0/1
  R (R1) 10.0.0.0
R (R3) 50.0.0.0
10.0.0.0/8
20.0.0.0/8
10.0.0.0/30
20.0.0.0/30

```

the process of sharing routing table is called convergence

Classful Routing

1. While sending periodic updates, it sends only Network Information, not the subnet mask info
 2. Does not support VLSM
 3. No ip subnet zero support
 4. eg. RIPv1 & IGRP
- IS, OSPF, BGP

Classless Routing

1. Sends both nw and subnet mask info
2. supports VLSM
3. Supports IP subnet
4. eg. RIPv2 EIGRP, IS-

200.0.0.0/26		255.255.255.192
X		X
0	64	128
1	65	193
62	126	
63	127	191
		255

=====

=====

convergence process of exchanging routing tables after every 30secs--- slow

=====

=

Routing Metrics: Routing Metric is a value used by a routing protocol to determine the best path to a destination network when multiple routes are available.

- | | |
|-------------------------------|--|
| 1. Hop Count | Number of routers a packet passes through |
| 2. Bandwidth | Link speed; higher bandwidth is preferred |
| 3. Delay of line destination) | Time taken for packet transmission(from source to destination) |
| 4. Cost | Administrative value assigned to links |
| 5. Reliability | Stability of a link 255/255--->reliable link |
| 6. Load | Current traffic utilization on a link,1/255-->less traffic,255/255-->100% traffic congestion |

7. MTU (Maximum Transmission Unit) Max Ethernet frame size can be 1518 bytes, that can be transmitted over a network without segmentation.

- | | |
|-----------------|--------------------------------------|
| 1. RIP | Hop Count |
| 2. IGRP & EIGRP | Bandwidth + Delay (Composite Metric) |
| 3. OSPF & IS-IS | Cost (based on bandwidth) |

=====

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RIP:

1. Routing Information Protocol.
2. Metrics: Hop Count
3. Maximum Hop count==15
4. small internetworks
5. Dynamic Routing Protocol
6. AD value 120
7. Works on Bellman Ford algorithm
8. RIPv1 & RIPv2
10. example of DVRP
11. slow convergence
12. 520/UDP
13. Supports Auto summarization
14. Periodic updates after every 30 seconds

RIPv2 Multicast Address---224.0.0.9

RIPv1	RIPv2
1. RIPv1 is a Distance-Vector Routing protocol. Vector Routing Protocol.	RIPv2 is also Distance-
2. It can supports class full network only. and classless networks.	It can support class full
3. It does not support for authentications.	It support for
4. It hop count limit is 15.	It hop count limit is 15.
5. Does not support for VLSM and discontinuous networks.	It supports for VLSM and
6. It is less secure.	It is more secure.
7. RIPv1 uses Broadcast traffic for updates.	RIPv2 uses Multicast
8. The routing update address used for Broadcast address used for Multicast is 224.0.0.9. is 255.255.255.255.	The routing update
9. RIPv1 does not provide trigger updates.	RIPv2 provides trigger
10. RIPv1 does not send a subnet mask to the routing table.	RIPv2 sends subnet mask
11. RIPv1 doesn't support manual route summarization.	RIPv2 supports manual
12. Does not support Classless Inter-Domain Routing (CIDR)	RIPv1 supports Classless

RIP has four timers update, invalid, hold down and flush timer,

1. Update Timer (30 seconds): The interval at which the router sends its full routing table to neighbors.
2. Invalid/Timeout Timer (180 seconds): If no update is received for a route within this time, the route is considered invalid, and the metric is set to 16.
3. Hold-down Timer (180 seconds): Prevents a router from accepting less favorable, unstable routes for a specific period after a route is marked invalid.
4. Flush Timer (240 seconds): The time before an invalid route is completely removed from the routing table

R1 R16 10 unreachable	R2 R50 10	R3 R100 10	R15 10
R1 -----> Routing table	R2 s1 ----->		s1 R3
c 70.0.0.0 f 50.0.0.0 f c 20.0.0.0 s0 40.0.0.0 s1 30 40 50	c 30.0.0.0 f c 20.0.0.0 s0 c 40.0.0.0 s1		c c
30.0.0.0	R1 10.0.0.0		R2
20.0.0.0	R3 50.0.0.0		R2
R2 user 10.0.02			R2 10.0.0.0

```
hold
10.0.0.2
user1 ping 50.0.0.2
```

```
R1          R2
R100
```

```
R15    R16
data    unreachable
```

RIPv2

First Router:

```
=====
#sh ip route
#conf t
config# router rip
      # network 192.168.1.0
      # network 192.168.1.80
      # version 2
      # do sh ip route
```

Middle Router:

```
=====
#sh ip route
#conf t
config# router rip
      # network 192.168.1.32
      # network 192.168.1.80
      # network 192.168.1.84
      # version 2
      # do sh ip route
```

Last Router:

```
=====
#sh ip route
#conf t
config# router rip
      # network 192.168.1.64
      # network 192.168.1.84
      # version 2
      # do sh ip route
```

```
=====
How to remove routing rip:
=====
```

```
config# no router rip
=====
```

Timers of RIP (seconds)

1. Route Update Timer	30
2. Invalid Timer	180
3. Hold-down Timer	180
4. Flush Timer	240